

Status. Verified-in-principle extension: the self-modulation entries $\mathbf{F}_e(4, 5) \ni (1 - \lambda_c)\delta\hat{x}_3$ and $\mathbf{H}(2, 5) \ni (\delta\hat{x}_3 - \delta\hat{x}_1)$ are not yet in the production controller (which uses the baseline $\mathbf{F}_e(4, 5) = \Delta h_d$, $\mathbf{H}(2, 5) = -\Delta H_d$; a'_x hold-unobservable, rank = 4). The rank $4 \rightarrow 5$ on-hold result requires these uncommitted entries.

Equation of motion

$$x[k+1] = x[k] + a_x[k] \{f_{dx}[k] + f_T[k]\} + \delta x_D[k]$$

Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k] \{f_{dx}[k] + f_T[k]\} - \delta x_D[k]$$

$$\delta x[k] = x_d[k] - x[k], \quad \Delta x_d[k] = x_d[k+1] - x_d[k].$$

Here f_T is the white thermal force, δx_D the disturbance contribution, and n_x the sensor noise (introduced below).

Gain via the actual height

($\delta h = -\delta x$ on the wall normal)

$$a_x[k] = a(h_d[k] + \delta h[k]) = a_{\det}[k] + a'_x \delta h[k] = a_{\det}[k] - a'_x \delta x[k]$$

$$a_{\det}[k] = a(h_d[k]), \quad a'_x = \left. \frac{da}{dh} \right|_{h_d[k]}$$

Measurement

(d -step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k]$$

Control objective

$$\delta x[k+1] = \lambda_c \delta x[k], \quad 0 \leq \lambda_c < 1$$

Ideal control effort (no delay), and the error it produces

$$\bar{f}_{dx}[k] = a_x^{-1}[k] \{\Delta x_d[k] + (1 - \lambda_c) \delta x[k] - \delta x_D[k]\}$$

$$\delta x[k+1] = \lambda_c \delta x[k] - a_x[k] f_T[k]$$

Current error over the d -step window

$$\delta x[k] = \delta x[k-d] + \sum_{i=1}^d \Delta x_d[k-i] - \sum_{i=1}^d a_x[k-i] \{f_{dx}[k-i] + f_T[k-i]\} - \sum_{i=1}^d \delta x_D[k-i]$$

Ideal control effort (d -step delay), and the error

$$\begin{aligned} \bar{f}_{dx}[k] = a_x^{-1}[k] \left\{ \underbrace{\left[\Delta x_d[k] + (1 - \lambda_c) \sum_{i=1}^d \Delta x_d[k-i] \right]}_{\Delta x_d^d[k]} + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] \right. \\ \left. - \underbrace{\left[\delta x_D[k] + (1 - \lambda_c) \sum_{i=1}^d \delta x_D[k-i] \right]}_{\delta x_D^d[k]} \right\} \end{aligned}$$

$$\delta x[k+1] = \lambda_c \delta x[k] - \underbrace{\left\{ a_x[k] f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_x[k-i] f_T[k-i] \right\}}_{\delta x_T^d[k]}$$

Implementable control effort

($\hat{a}_x$ for a_x , δx_m for $\delta x[k-d]$, disturbance dropped)

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k] \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x_m[k] - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}}$$

$$\delta x[k+1] = \lambda_c \delta x[k] - a_x[k] [f_{dx}[k] - \bar{f}_{dx}[k]] - \delta x_T^d[k]$$

$$a_x[k] \bar{f}_{dx}[k] = \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] - \delta x_D^d[k]$$

With the gain error $e_{a_x} = a_x - \hat{a}_x$,

$$\begin{aligned} a_x[k] f_{dx}[k] = (1 + e_{a_x}[k] \hat{a}_x^{-1}[k]) \left\{ \Delta x_d^d[k] \right. \\ \left. + (1 - \lambda_c) \left[(\delta x[k-d] + n_x[k]) - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}, \end{aligned}$$

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{f}_{dx}[k] = e_{a_x}[k] f_{dx}[k] + (1 - \lambda_c) n_x[k] \\ + (1 - \lambda_c) \sum_{i=1}^d (a_x[k-i] - \hat{a}_x[k-i]) f_{dx}[k-i] + \delta x_D^d[k]. \end{aligned}$$

Parameter model

$$a_x[k] = a_{\det}[k] + a_{xram}[k], \quad a_{xram}[k] = a'_x x_{ram}[k] = -a'_x \delta x[k]$$

$$\boxed{a_x[k+1] - a_x[k] = a'_x (\Delta h_d[k] - \Delta \delta x[k]), \quad \Delta h_d[k] = h_d[k+1] - h_d[k]}$$

Close the loop ($\delta x_3 = \delta x[k]$, the current error; ε is the lumped noise defined on the next page):

$$\delta x[k+1] = \lambda_c \delta x_3[k] - \varepsilon[k] \Rightarrow \Delta \delta x[k] = (\lambda_c - 1) \delta x_3[k] - \varepsilon[k] = -(1 - \lambda_c) \delta x_3[k] - \varepsilon[k]$$

$$a_x[k+1] = \underbrace{a_x[k]}_{\mathbf{F}_e(4,4)=1} + \underbrace{a'_x [\Delta h_d[k] + (1 - \lambda_c) \delta x_3[k]]}_{\text{known + self-dither}} + \underbrace{a'_x \varepsilon[k]}_{w_4[k] \rightarrow q_4}$$

$$\begin{cases} a_x[k+1] = a_x[k] + a'_x [\Delta h_d[k] + (1 - \lambda_c) \delta x_3[k]] + w_4[k] \\ a'_x[k+1] = a'_x[k] + w_{a'}[k] \end{cases}$$

Estimator (predict at $\delta \hat{x}_3$), errors, error dynamics

$$\hat{a}_x[k+1] = \hat{a}_x[k] + \hat{a}'_x[k] [\Delta h_d[k] + (1 - \lambda_c) \delta \hat{x}_3[k]], \quad \hat{a}'_x[k+1] = \hat{a}'_x[k]$$

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k], \quad e_{a'}[k] = a'_x[k] - \hat{a}'_x[k]$$

$$\begin{cases} e_{a_x}[k+1] = e_{a_x}[k] + [\Delta h_d[k] + (1 - \lambda_c) \delta \hat{x}_3[k]] e_{a'}[k] + w_4[k] \\ e_{a'}[k+1] = e_{a'}[k] + w_{a'}[k] \end{cases}$$

Gain error over the window ($e_{a'}$ const; $h_d[k] - h_d[k-i] = \sum_{j=1}^i \Delta h_d[k-j]$)

$$a_x[k-i] - \hat{a}_x[k-i] \approx e_{a_x}[k] - (h_d[k] - h_d[k-i]) e_{a'}[k] - \sum_{j=1}^i \delta a_{ram}[k-j], \quad \delta a_{ram} = a'_x \Delta x_{ram}$$

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{f}_{dx}[k] &\approx e_{a_x}[k] \underbrace{\left\{ f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k-i] \right\}}_{F_{dx}[k]} \\ &\quad - e_{a'}[k] \underbrace{\left\{ (1 - \lambda_c) \sum_{i=1}^d (h_d[k] - h_d[k-i]) f_{dx}[k-i] \right\}}_{dF_{dx}^h[k]} \\ &\quad - \underbrace{\left\{ (1 - \lambda_c) \sum_{i=1}^d (d+1-i) f_{dx}[k-i] \delta a_{ram}[k-i] \right\}}_{\delta x_{\delta a_f}^d[k]} \\ &\quad + \delta x_D^d[k] + (1 - \lambda_c) n_x[k] \end{aligned}$$

$$\delta x[k+1] = \lambda_c \delta x[k] - F_{dx}[k] e_{a_x}[k] + dF_{dx}^h[k] e_{a'}[k] - \delta x_D^d[k] - \underbrace{\left\{ \delta x_T^d[k] - \delta x_{\delta a_f}^d[k] + (1 - \lambda_c) n_x[k] \right\}}_{\varepsilon[k]}$$

State equation

($\delta x_D^d \equiv 0$; error row at baseline, route A)

$$\delta x_1[k+1] = \delta x_2[k], \quad \delta x_2[k+1] = \delta x_3[k],$$

$$\delta x_3[k+1] = \lambda_c \delta x_3[k] - F_{dx}[k] e_{a_x}[k] + dF_{dx}^h[k] e_{a'}[k] - \varepsilon[k],$$

$$a_x[k+1] = a_x[k] + a'_x [\Delta h_d[k] + (1 - \lambda_c) \delta x_3[k]] + w_4[k],$$

$$a'_x[k+1] = a'_x[k] + w_{a'}[k].$$

Output equation

$$a_x[k-d] = a_x[k] - a'_x[\Delta H_d[k] - (\delta x_3[k] - \delta x_1[k])], \quad \Delta H_d[k] = h_d[k] - h_d[k-d]$$

with n_a the gain-measurement noise:

$$\begin{aligned} y_1[k] &= \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]}, \\ y_2[k] &= a_{xm}[k] = a_x[k-d] + n_a[k] \\ &\approx a_x[k] - a'_x[\Delta H_d[k] - (\delta x_3[k] - \delta x_1[k])] + \underbrace{n_a[k]}_{r_2[k]}. \end{aligned}$$

The whole span $a'_x(\delta x_3 - \delta x_1)$ is now a deterministic-in-state term carried by $\mathbf{H}$, so the only true measurement noise is n_a . (The baseline used $\mathbf{H}(2, 5) = -\Delta H_d$ and absorbed the unmodelled span into $r_2 = n_a - \sum_i \delta a_{ram}[k-i]$; since $\sum_i \delta a_{ram}[k-i] \simeq -a'_x(\delta x_3 - \delta x_1)$, keeping that r_2 here would count the span once as the $\mathbf{H}(2, 5)$ signal and again as noise – a double-count that inflates R_{22} .)

$$\boxed{\mathbf{H}(2, 5) = -\Delta H_d[k] + (\delta \hat{x}_3[k] - \delta \hat{x}_1[k])}$$

$$\mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -\hat{a}'_x & 0 & \hat{a}'_x & 1 & -\Delta H_d[k] + (\delta \hat{x}_3 - \delta \hat{x}_1) \end{bmatrix}, \quad \mathbf{y}[k] = \mathbf{H}[k] \mathbf{x}[k] + \mathbf{r}[k]$$

($\mathbf{H}(2, 1) = -\hat{a}'_x$, $\mathbf{H}(2, 3) = +\hat{a}'_x$ are $O(\hat{a}'_x)$, negligible; the load-bearing new term is $\mathbf{H}(2, 5)$.)

Closed-loop estimator

$$e_{\delta x_1} = \delta x_1 - \delta \hat{x}_1, \quad e_{a_x} = a_x - \hat{a}_x, \quad e_{a'} = a'_x - \hat{a}'_x$$

$$e_{y_1}[k] = \delta x_m[k] - \delta \hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$\begin{aligned} e_{y_2}[k] &= a_{xm}[k] - \hat{a}_x[k] + [\Delta H_d[k] - (\delta \hat{x}_3 - \delta \hat{x}_1)] \hat{a}'_x[k] \\ &= e_{a_x}[k] - [\Delta H_d[k] - (\delta \hat{x}_3 - \delta \hat{x}_1)] e_{a'}[k] + \underbrace{n_a[k] + \hat{a}'_x(e_{\delta x_3} - e_{\delta x_1})}_{r_2[k] \approx n_a} \end{aligned}$$

$$\delta \hat{x}_1[k+1] = \delta \hat{x}_2[k] + \ell_{11} e_{y_1} + \ell_{12} e_{y_2},$$

$$\delta \hat{x}_2[k+1] = \delta \hat{x}_3[k] + \ell_{21} e_{y_1} + \ell_{22} e_{y_2},$$

$$\delta \hat{x}_3[k+1] = \lambda_c \delta \hat{x}_3[k] + \ell_{31} e_{y_1} + \ell_{32} e_{y_2},$$

$$\hat{a}_x[k+1] = \hat{a}_x[k] + \hat{a}'_x[k] [\Delta h_d[k] + (1 - \lambda_c) \delta \hat{x}_3[k]] + \ell_{41} e_{y_1} + \ell_{42} e_{y_2},$$

$$\hat{a}'_x[k+1] = \hat{a}'_x[k] + \ell_{51} e_{y_1} + \ell_{52} e_{y_2}.$$

Error dynamics

$$\begin{aligned}
G_4[k] &:= \Delta h_d[k] + (1 - \lambda_c) \delta \hat{x}_3[k] \quad (\mathbf{F}_e(4, 5)), \quad G_H[k] := \Delta H_d[k] - (\delta \hat{x}_3 - \delta \hat{x}_1) \quad (-\mathbf{H}(2, 5)) \\
e_{\delta x_1}[k+1] &= e_{\delta x_2} - \ell_{11} e_{\delta x_1} - \ell_{12} e_{a_x} + \ell_{12} G_H e_{a'}, \\
e_{\delta x_2}[k+1] &= e_{\delta x_3} - \ell_{21} e_{\delta x_1} - \ell_{22} e_{a_x} + \ell_{22} G_H e_{a'}, \\
e_{\delta x_3}[k+1] &= \lambda_c e_{\delta x_3} - F_{dx} e_{a_x} + dF_{dx}^h e_{a'} - \ell_{31} e_{\delta x_1} - \ell_{32} e_{a_x} + \ell_{32} G_H e_{a'} - \underbrace{\varepsilon[k]}_{q_3}, \\
e_{a_x}[k+1] &= e_{a_x} + G_4 e_{a'} - \ell_{41} e_{\delta x_1} - \ell_{42} e_{a_x} + \ell_{42} G_H e_{a'} + \underbrace{w_4[k]}_{q_4}, \\
e_{a'}[k+1] &= e_{a'} - \ell_{51} e_{\delta x_1} - \ell_{52} e_{a_x} + \ell_{52} G_H e_{a'} + \underbrace{w_{a'}[k]}_{q_5}.
\end{aligned}$$

Error-dynamics matrix

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & dF_{dx}^h[k] \\ 0 & 0 & 0 & 1 & \Delta h_d[k] + (1 - \lambda_c) \delta \hat{x}_3[k] \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k-i], \quad dF_{dx}^h[k] = (1 - \lambda_c) \sum_{i=1}^d (h_d[k] - h_d[k-i]) f_{dx}[k-i]$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H}[k] \mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}[k] \mathbf{r}[k]$$

Change versus the baseline slope estimator (5state_est_aprime)

$\mathbf{F}_e(4, 5) = \Delta h_d[k] + (1 - \lambda_c) \delta \hat{x}_3[k]$	(baseline $\Delta h_d[k]$)
$\mathbf{H}(2, 5) = -\Delta H_d[k] + (\delta \hat{x}_3[k] - \delta \hat{x}_1[k])$	(baseline $-\Delta H_d[k]$)

On a hold $\Delta h_d = \Delta H_d = dF_{dx}^h = 0$: column 5 collapses to the thermal terms $(1 - \lambda_c) \delta \hat{x}_3$ and $\delta \hat{x}_3 - \delta \hat{x}_1$ (Page 7). Route B (cross-check, not additive): substituting $a_x = a_{\text{det}} - a'_x \delta x$ into the δx_3 row instead gives $\mathbf{F}_e(3, 5) += \delta \hat{x}_3 f_{dx}$, $\mathbf{F}_e(3, 3) = \lambda_c + \hat{a}'_x f_{dx}$; equivalent at first order, same Fisher information. Route A is implemented.

Gain-state model: reconciliation with the 4-state AR(1)

The 4-state (known wall, 4state_del_hd) noted that the gain fluctuation $a_{x_{ram}} = -a'_x \delta x$ is a bounded AR(1) process, not a random walk. With $\delta x[k+1] = \lambda_c \delta x[k] - \varepsilon[k]$,

$$a_{x_{ram}}[k+1] = -a'_x \delta x[k+1] = \lambda_c a_{x_{ram}}[k] + a'_x \varepsilon[k] \quad (\text{pole } \lambda_c).$$

Why the 4-state's $\mathbf{F}_e(4, 4) = \lambda_c$ does not carry over

In the 4-state the wall is known, so a_{det} is a signal and $a_x = a_{\text{det}} + a_{x_{\text{ram}}}$ with $a_{x_{\text{ram}}} = a_x - a_{\text{det}}$. The AR(1) reversion is then a multiple of the state:

$$a_x[k+1] = \underbrace{\lambda_c a_x[k]}_{\mathbf{F}_e(4,4)=\lambda_c} + (1 - \lambda_c) a_{\text{det}}[k] + \Delta a_x[k] + a'_x \varepsilon[k] \quad (4\text{-state, known } a_{\text{det}}).$$

In the unknown-wall 5-state there is no a_{det} signal (avoiding $c(\bar{h})$ is the reason for estimating a'_x). With no reversion target, $(1 - \lambda_c)a_{\text{det}}$ cannot be formed and the reversion cannot live in $\mathbf{F}_e(4, 4)$. The gain is a free integrated state:

$\mathbf{F}_e(4, 4) = 1 \quad (\text{unknown wall: free } a_x; \text{ the 4-state's } \lambda_c \text{ needs the known } a_{\text{det}} \text{ anchor}).$

This corrects a cross-document slip: `5state_aprime_observability.tex` imported $\mathbf{F}_e(4, 4) = \lambda_c$; the correct value here is 1, matching `build_F_e_5state_aprime.m` (row 4 = `[0001*]`).

Where the AR(1) physics goes instead

The reversion does not vanish; it relocates to three places inherited from the 4-state.

First, into the $(1 - \lambda_c)$ factor in $\mathbf{F}_e(4, 5)$. From Page 2, $\Delta \delta x = -(1 - \lambda_c)\delta \hat{x}_3 - \varepsilon$, so the deterministic self-dither carries the closed-loop pole as its coefficient,

$$\mathbf{F}_e(4, 5) = \Delta h_d + (1 - \lambda_c) \delta \hat{x}_3 = \left. \frac{\partial a_x[k+1]}{\partial a'_x} \right|_{\delta \hat{x}},$$

where $(1 - \lambda_c)$ is the AR(1) reversion factor (one minus the pole λ_c). The 4-state hides the pole in $\mathbf{F}_e(4, 4) = \lambda_c$; the 5-state exposes it as the coefficient of $\delta \hat{x}_3$.

Second, into Q_{44} , the AR(1) white innovation $w_4 = a'_x \varepsilon$ (distinct from the realized increment δa_{ram} below). With $\text{Var}(\varepsilon) = (1 - \lambda_c^2) \text{Var}(\delta x)$ and $\text{Var}(\delta x) = C_{\delta x} \sigma_{\delta h}^2$, $C_{\delta x} = 2 + \frac{1}{1 - \lambda_c^2}$,

$q_4 = \text{Var}(a'_x \varepsilon) = a_x'^2 (1 - \lambda_c^2) C_{\delta x} \sigma_{\delta h}^2 = (3 - 2\lambda_c^2) a_x'^2 \sigma_{\delta h}^2, \quad \sigma_{\delta h}^2 = 4k_B T a_x,$

identical to the 4-state `use_q44_ar1` driver, using $a'_x = -a_x K_h / R$.

Third, into the conflation split. The white driver q_4 is not the realized increment $\delta a_{\text{ram}} = a'_x \Delta x_{\text{ram}}$. The realized increment feeds Q_{33} (the δx_3 random-gain term); the white driver feeds Q_{44} :

$$\underbrace{\text{Var}(\delta a_{\text{ram}}) = \frac{2}{1 + \lambda_c} a_x'^2 \sigma_{\delta h}^2}_{\text{var_da_inc} \rightarrow Q_{33}} \quad \text{vs} \quad \underbrace{q_4 = (3 - 2\lambda_c^2) a_x'^2 \sigma_{\delta h}^2}_{\text{var_da_ram} \rightarrow Q_{44}}.$$

Using one number for both double-counts the physics (the 4-state's conflation bug). One difference from the 4-state: the realized increment does **not** feed R_{22} here. The baseline routed the d -step span into R_{22} (the $-\sum \delta a_{\text{ram}}$ term in r_2) because $\mathbf{H}$ did not model it; the self-mod carries that span in $\mathbf{H}(2, 5)$, so $R_{22} = \text{Var}(n_a)$ (the a_{xm} floor) alone (Output equation, Page 3). Together, $\{\mathbf{F}_e(4, 4) = 1, \mathbf{F}_e(4, 5) \ni (1 - \lambda_c)\delta \hat{x}_3, q_4, \text{split}\}$ *aims to* reproduce the 4-state's gain-level tracking while exposing the slope – but the gain-level reversion is weak without a known a_{det} (Caveat 6).

Observability on a height hold

On a hold $\Delta h_d = \Delta H_d = dF_{dx}^h = 0$, and column 5 collapses to its thermal terms:

$$\mathbf{F}_e[k] \mathbf{e}_5 \Big|_{\text{hold}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ (1 - \lambda_c) \delta \hat{x}_3 \\ 1 \end{bmatrix}, \quad \mathbf{H}[k] \mathbf{e}_5 \Big|_{\text{hold}} = \begin{bmatrix} 0 \\ \delta \hat{x}_3 - \delta \hat{x}_1 \end{bmatrix}.$$

The first block row of $\mathcal{O}_N$ already projects onto $\mathbf{e}_5$ through a_{xm} :

$$\mathcal{O}_N(k) \mathbf{e}_5 = \begin{bmatrix} \mathbf{H}[k] \mathbf{e}_5 \\ \vdots \end{bmatrix} = \begin{bmatrix} 0 \\ \delta \hat{x}_3[k] - \delta \hat{x}_1[k] \\ \vdots \end{bmatrix} \neq \mathbf{0} \quad (\delta \hat{x}_3 \neq \delta \hat{x}_1).$$

Baseline: this entry was $-\Delta H_d = 0$ on a hold, so $\text{rank} = 4$, $\ker \mathcal{O}_N = \text{span}\{\mathbf{e}_5\}$ (slope the lone unobservable mode). With the self-dither restored:

$$\boxed{\text{rank } \mathcal{O}_N = 5 \text{ on a hold} \iff \delta \hat{x}_3 - \delta \hat{x}_1 \neq 0 \quad (\Delta h_d = 0).}$$

The gain is sampled at two slightly different actual heights $h_d + \delta x[k]$ and $h_d + \delta x[k-d]$; their difference exposes the slope, with no commanded motion.

Rank is necessary, not sufficient

The excitation is the thermal span $|\delta \hat{x}_3 - \delta \hat{x}_1| \sim \text{tens of nm}$ ($\sigma_{\delta x} \approx 16\text{--}24 \text{ nm}$), versus μm for a commanded sweep:

$$\boxed{\begin{aligned} \|\mathcal{O}_N \mathbf{e}_5\|_{\text{hold}} &\sim |\delta \hat{x}_3 - \delta \hat{x}_1| \ll |\Delta h_d|, \\ \frac{\sigma_5}{\sigma_1} &\sim 6.4 \times 10^{-3} \text{ (10 nm span, } N=8 \text{ window, linear in span).} \end{aligned}}$$

The binding limit on a hold is the self-dither SNR, beneath the gain noise $a_x f_T$ (Q_{33}) and the a_{xm} chi-squared floor ($R_{22} \sim 42\%$). A commanded $\Delta h_d \neq 0$ remains the strong, reliable excitation; the self-modulation is the weak floor that keeps a'_x off the null space on a hold.

Caveats and failure modes

1. **Errors-in-variables.** The self-dither regressor ($\delta \hat{x}_3, \delta \hat{x}_3 - \delta \hat{x}_1$) is the filter's own noisy estimate, so the learned slope is biased toward zero by the reliability ratio $\text{Var}(\delta x) / (\text{Var}(\delta x) + \text{Var}(e_{\delta x})) < 1$. Worst on a hold, where the whole signal is this regressor; absent for a noise-free commanded sweep.
2. **Common thermal noise (largely removed by the r_2 correction).** In the baseline accounting the gain-measurement noise carried $-\sum \delta a_{ram}[k-i] \simeq +a'_x(\delta x_3 - \delta x_1)$ – the same span $\mathbf{H}(2, 5)$ now models – an exact first-order double-count (signal in $\mathbf{H}$, noise in R_{22}). It is removed structurally by setting $r_2 = n_a$ (Output equation), not by an $\mathbf{R}$ off-diagonal. What

remains is the smaller second-order errors-in-variables residual $\hat{a}'_x(e_{\delta x_3} - e_{\delta x_1})$ plus the genuine correlation between a_{xm} 's intrinsic chi-squared noise and the δx regressor; these still bias a pure-hold estimate and need an $\mathbf{R}$ off-diagonal or a whitened residual.

3. **Near-wall gain-measurement bias.** The a_{xm} chain's C_{dpmr} degrades when $\hat{a}_x \neq a_x$; being height-dependent it masquerades as a slope and compounds the previous item. The central near-wall difficulty, carried from `5state_est_aprime`.
4. **Wall-normal axis only.** $\delta h = -\delta x$ holds only on the normal axis; on x, y the height does not move with the tracking error ($\partial a_x / \partial \delta x = 0$), the self-mod column is zero, and tangential a'_x stays hold-unobservable.
5. **First-order filter vs. UKF.** The self-modulation is bilinear ($a'_x \delta x f_{dx}$); the Jacobians linearize at $\delta \hat{x}$ and miss $\text{Cov}(a'_x, \delta x)$. On a hold the whole signal is that second-order term, so a UKF / sigma-point (or iterated / 2nd-order EKF) is materially better.
6. **No gain-level reversion near the wall.** $\mathbf{F}_e(4, 4) = 1$ is a free integrator. Unlike the 4-state (which reverts to the *known* a_{det} , an $O(a_{\text{nom}})$ pull), the unknown-wall gain LEVEL has only the zero-mean $O(\delta x)$ self-dither ($\sim$ tens of nm, $\sim 10^2$ – $10^3 \times$ weaker) and the a_{xm} measurement to bound it. When the near-wall gate turns a_{xm} off ($\bar{h} < \bar{h}_{\text{safe}}$), a_x has neither reversion nor measurement and drifts as an undamped random walk – the failure the 4-state $\mathbf{F}_e(4, 4) = \lambda_c$ was introduced to cure. A state-computable reversion ($a_{x_{ram}} = -a'_x \delta x_3$) or a near-wall gate-off stability check is required before closing the loop; “aims to reproduce 4-state tracking” (Page 6) is unverified for the unknown-wall case.

Bottom line. Restoring the thermal $-a'_x \delta x$ coupling moves a'_x from structurally unobservable to structurally observable on a hold; practical estimability is then an SNR question, eroded by items 1–2, with a commanded $\Delta h_d \neq 0$ the reliable excitation.